



MAKE DATA-DRIVEN POINTS

Our scientific arguments
must be structured like scaffolding

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MASSHINE

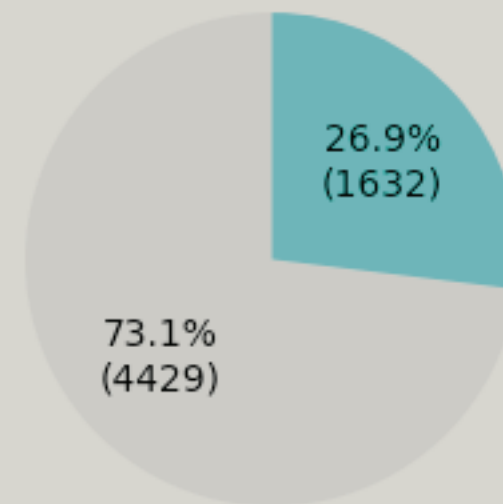
Summary

1. Measurements
2. Circulating reference
3. Truth and robustness
4. The Toulmin approach to building an argument
5. How to use NotebookLM to explore potential arguments
6. What is retrieval augmented generation (RAG)?

Measurements

The visualizations and numbers we obtain from the dataset through the notebooks constitute *measurements*.

Statements mentioning 'bornholm' (case-insensitive)

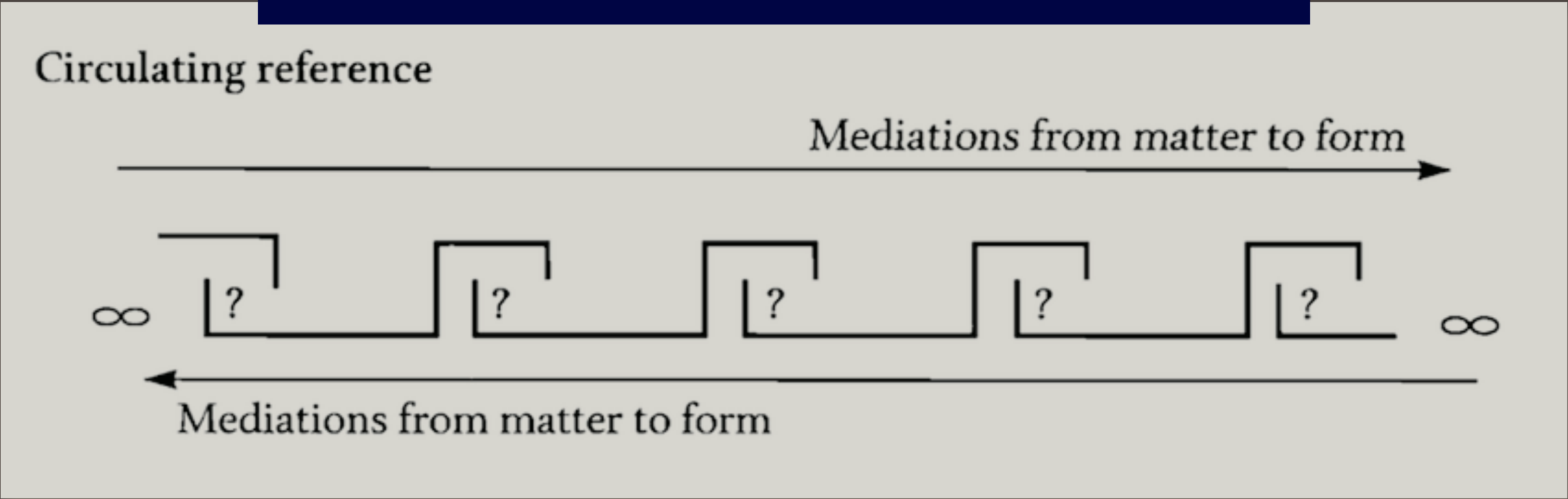


This visualization *measures* that 26.9% of the corpus' statements mention “Bornholm” (in a case-insensitive way).

This number does not depend on the reader's subjectivity.

The circulating reference

Scientific facts gain their power to represent the world by refering to it through chains of transformations where each step changes the format but maintains the connection.



Yes, that guy again...



*Phenomenon
(public space)*

Dataset

*Data
visualization*

Measurement

*Argument
(policy brief)*

Truth and robustness



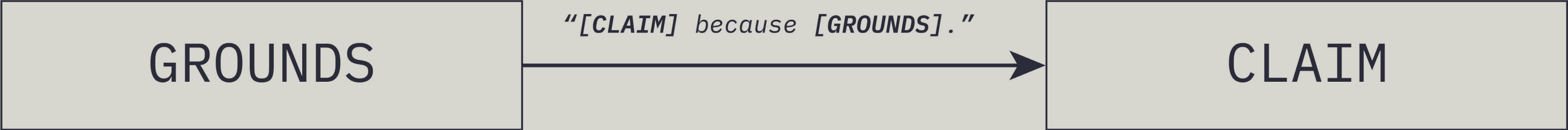
Robust arguments have a specific structure. It allows your audience following your rationale: they can understand what evidence you're using and how you're building your case from it.

Yes, that guy again...



The Toulmin approach to building an argument

Is this how you think
of an argument?



The Toulmin approach to building an argument

Is this how you think
of an argument?

GROUND

During a lunar eclipse, you can see that Earth's shadow projected onto the Moon is round.

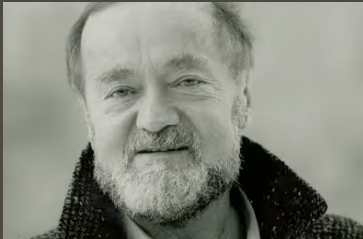
*"The Earth is round because,
as you can see during a lunar
eclipse, its projected shadow
onto the Moon is round."*

CLAIM

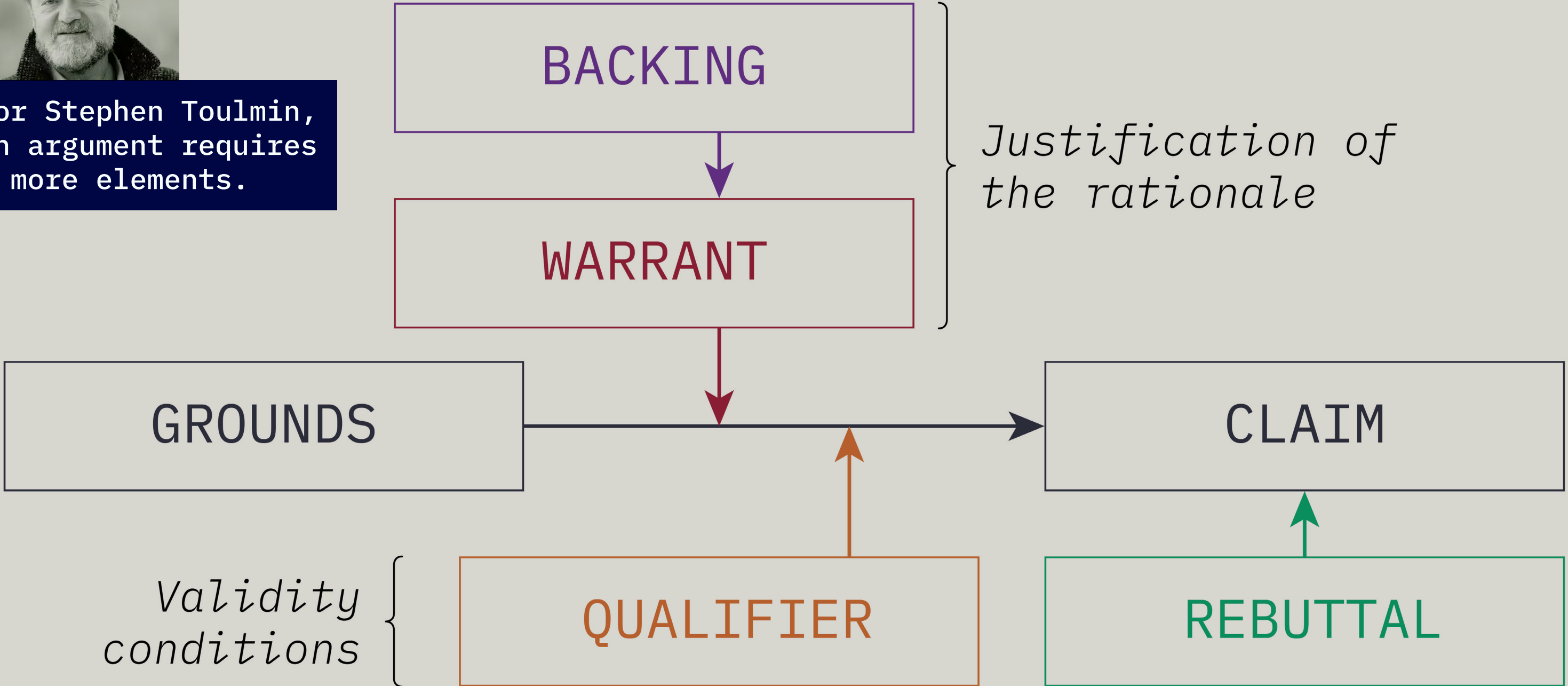
The Earth is round.

The problem is the
mediation from the
ground(s) to the claim.
It is not self-evident.

The Toulmin approach to building an argument



For Stephen Toulmin,
an argument requires
4 more elements.



The Toulmin approach to building an argument

BACKING

Support or explanation provided for the warrant. Fact-like. Many kinds are possible depending on the field.

“[**GROUND**S]. And we know that [**WARRANT**], because [**BACKING**]. So [**QUALIFIER**] [**CLAIM**], unless [**REBUTTAL**].”

WARRANT

Bridge-like statement explaining why the claim follows from the grounds. General rule justifying the claim given the grounds.

The *warrant*, in particular, is really important.

GROUNDS

Backing for the claim: data and measurement (among other possibilities).

CLAIM

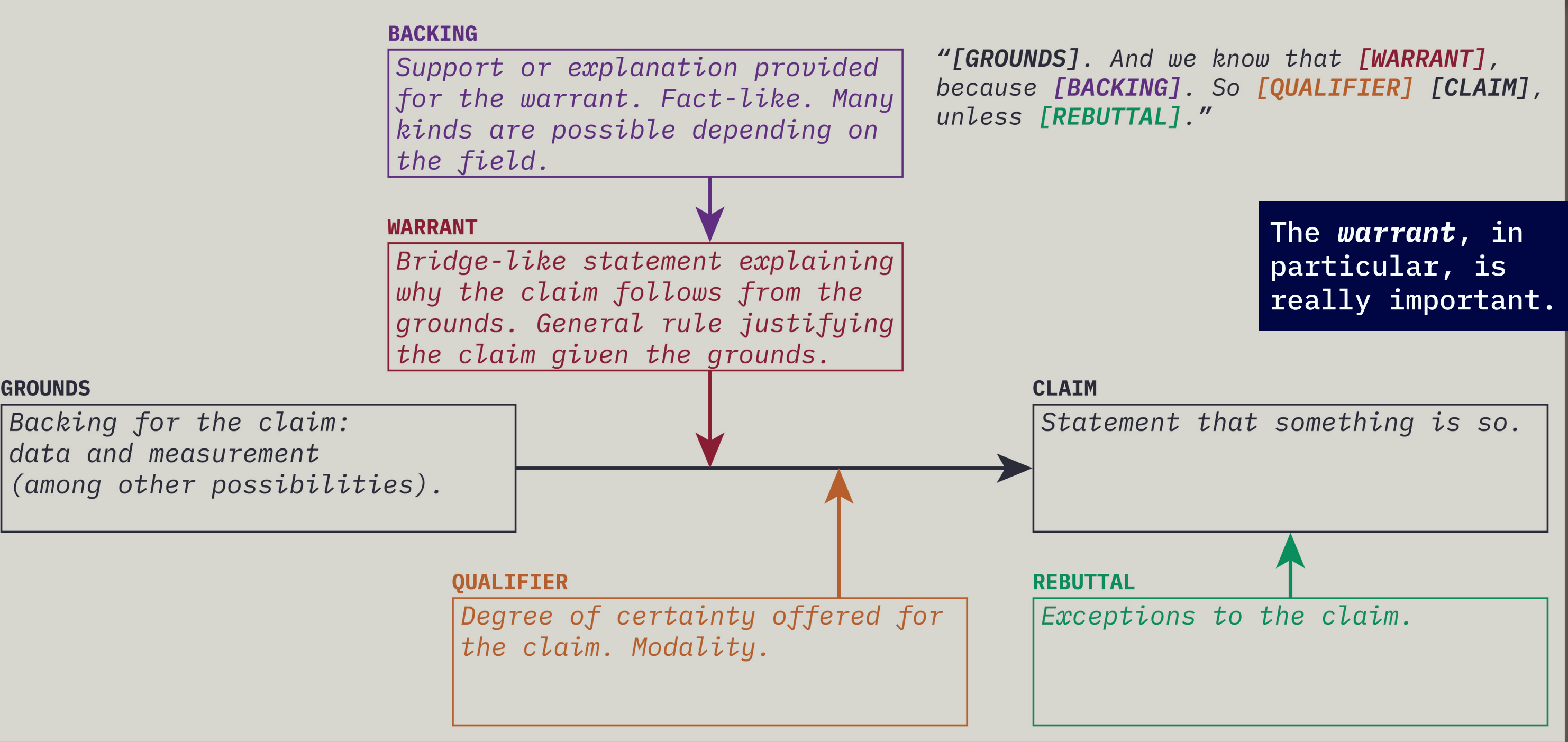
Statement that something is so.

QUALIFIER

Degree of certainty offered for the claim. Modality.

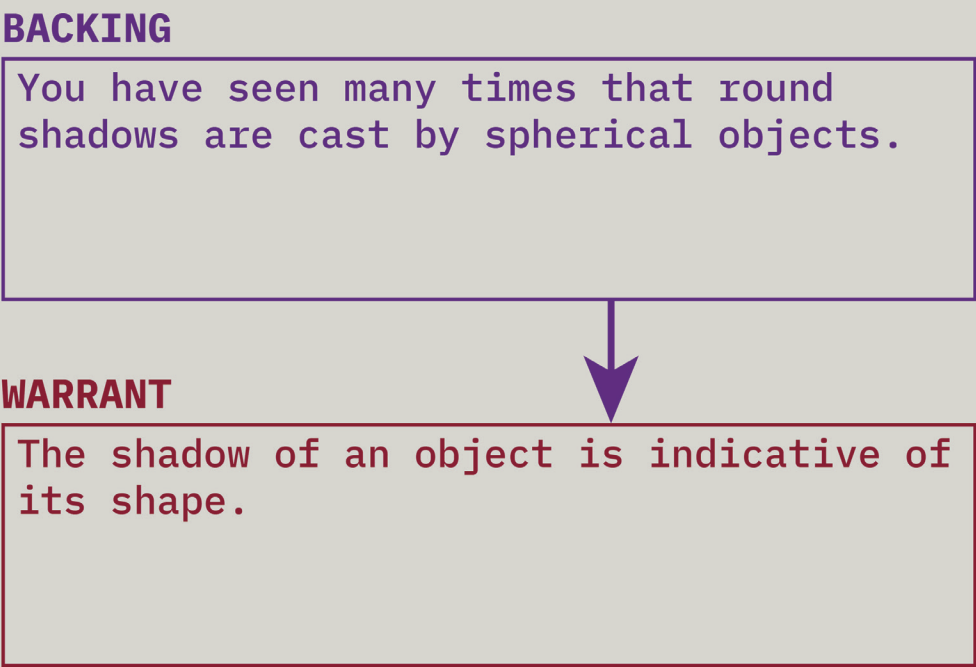
REBUTTAL

Exceptions to the claim.



The Toulmin approach to building an argument

Example 1:
Earth is round



GROUND

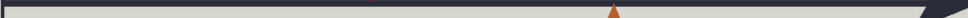
During a lunar eclipse, you can see that Earth's shadow projected onto the Moon is round.

CLAIM

The Earth is round.

QUALIFIER

You can directly see that...



REBUTTAL

Unless the Earth were shaped like a disc or another non-spherical shape that casts a round shadow.



"During a lunar eclipse, you can see that Earth's shadow projected onto the Moon is round. And you know that the shadow of an object is indicative of its shape, because that is something you have seen many times. So you can directly see that the Earth is round, unless it were shaped like a disc or another non-spherical shape that casts a round shadow."

The Toulmin approach to building an argument

Example 2:
Importance of Bornholm
to the Energy Islands
project

BACKING

The corpus captures the voice many different actors in many different spaces about Energy Islands.

WARRANT

Terms recurring in this corpus are important elements of the discourses about the Energy Islands project.

GROUND

1632 of 6061 actor statements (26.9%) mention the word "Bornholm".

CLAIM

Bornholm is an important part of the Energy Islands project.

QUALIFIER

So we can reasonably consider that...

REBUTTAL

Unless the corpus were stronlgy biased in favor of mentioning Bornholm.

"1632 of 6061 actor statements (26.9%) mention the word 'Bornholm'. And we know that *terms recurring in this corpus are important elements of the discourses about the Energy Islands project*, because the corpus captures the voice many different actors in many different spaces about it. So *we can reasonably consider that Bornholm is an important part of the Energy Islands project*, unless *the corpus were stronlgy biased in that favor.*"

NotebookLM is a playground
where you can experiment
with building arguments
about the dataset.

Ingest dataset into *NotebookLM*

Tech demo

RAG is how
NotebookLM works
under the hood

What is RAG? (retrieval augmenter generation)

Show & Tell with Nomic Atlas



THANK YOU FOR YOUR ATTENTION

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MACHINE